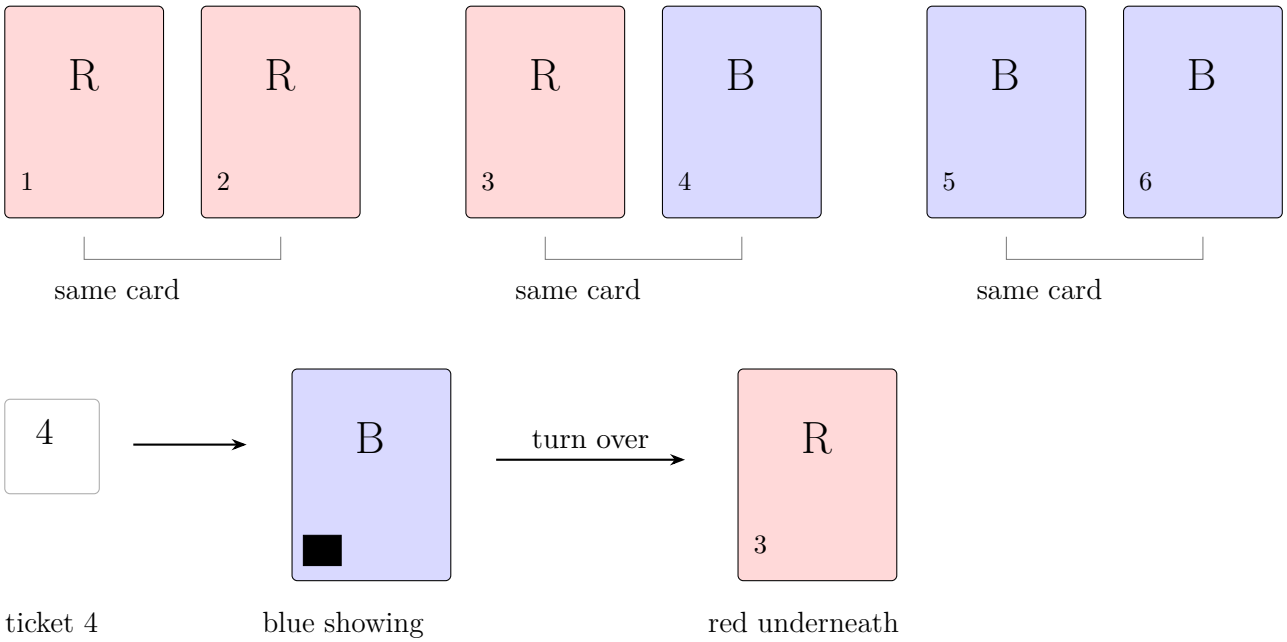
The small numbers match the six tickets. A secret ticket picks the face that shows, with its number covered. You see only the color.



Problem 1: Sort every ticket by its hidden color. Then find two tickets that show the same color but hide different colors.

red underneath

blue underneath

Problem 2: Red is showing; which hidden color has more ways? Try removing one ticket, and find every removal that makes a tie.

red underneath

blue underneath

Problem 3: Put all six tickets back; now blue is showing. Find every one-ticket removal that gives a tie between the hidden colors.

red underneath

blue underneath

Only tickets 1 and 3 are in the cup.



Problem 4: Add just one extra ticket to this cup: 2, 4, 5, or 6. Among red-showing rounds, which hidden color has more ways before and after each addition?

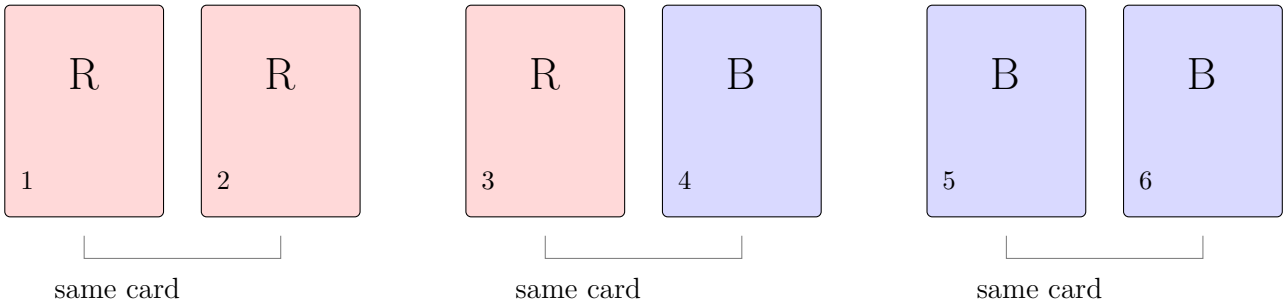
red underneath

blue underneath

Problem 5: Choose three tickets so red sometimes shows and then always hides blue. Find every choice that works.

Use all six face tickets again. Lay each card's two tickets on it, so removing a card takes both.

Problem 6: Remove one card so red showing always hides red. Can you remove a different card so red showing always hides blue?



Problem 7: Find every two-ticket cup where red showing gives a tie between the hidden colors. Red must sometimes show.